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1 "D:\Poojangi\Research Paper\GA for TSP\venv\Scripts\
python.exe" "D:/Poojangi/Research Paper/GA for TSP/
testing2.py"
2 Iteration: 0
3 The best route so far is: [0, 1, 10, 13, 3, 9, 2, 11
, 12, 7, 14, 6, 4, 8, 5] with distance: 579.0
4 -----
5 Iteration: 1
6 The best route so far is: [0, 1, 10, 13, 3, 9, 2, 11
, 12, 7, 14, 6, 4, 8, 5] with distance: 579.0
7 -----
8 Iteration: 2
9 The best route so far is: [0, 1, 10, 13, 3, 9, 2, 11
, 12, 7, 14, 6, 4, 8, 5] with distance: 579.0
10 -----
11 Iteration: 3
12 The best route so far is: [3, 9, 2, 11, 7, 14, 4, 13
, 10, 12, 6, 1, 8, 5, 0] with distance: 575.0
13 -----
14 Iteration: 4
15 The best route so far is: [3, 9, 2, 11, 7, 14, 4, 13
, 10, 12, 6, 1, 8, 5, 0] with distance: 575.0
16 -----
17 Iteration: 5
18 The best route so far is: [3, 9, 2, 11, 7, 14, 4, 13
, 10, 12, 6, 1, 8, 5, 0] with distance: 575.0
19 -----
20 Iteration: 6
21 The best route so far is: [0, 14, 4, 5, 9, 10, 3, 7
, 13, 11, 1, 2, 6, 8, 12] with distance: 523.0
22 -----
23 Iteration: 7
24 The best route so far is: [1, 8, 4, 7, 9, 10, 5, 11
, 3, 12, 14, 6, 2, 13, 0] with distance: 516.0
25 -----
26 Iteration: 8
27 The best route so far is: [1, 8, 4, 7, 9, 10, 5, 11
, 3, 12, 14, 6, 2, 13, 0] with distance: 516.0
28 -----
29 Iteration: 9
30 The best route so far is: [1, 8, 4, 7, 9, 10, 5, 11
, 3, 12, 14, 6, 2, 13, 0] with distance: 516.0
31 -----
32 Iteration: 10
```

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33 The best route so far is: [1, 8, 4, 7, 9, 10, 5, 11
    , 3, 12, 14, 6, 2, 13, 0] with distance: 516.0
34 -----
35 Iteration: 11
36 The best route so far is: [1, 8, 4, 7, 9, 10, 5, 11
    , 3, 12, 14, 6, 2, 13, 0] with distance: 516.0
37 -----
38 Iteration: 12
39 The best route so far is: [1, 12, 2, 13, 5, 7, 9, 11
    , 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
40 -----
41 Iteration: 13
42 The best route so far is: [1, 12, 2, 13, 5, 7, 9, 11
    , 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
43 -----
44 Iteration: 14
45 The best route so far is: [1, 12, 2, 13, 5, 7, 9, 11
    , 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
46 -----
47 Iteration: 15
48 The best route so far is: [1, 12, 2, 13, 5, 7, 9, 11
    , 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
49 -----
50 Iteration: 16
51 The best route so far is: [1, 12, 2, 13, 5, 7, 9, 11
    , 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
52 -----
53 Iteration: 17
54 The best route so far is: [1, 12, 2, 13, 5, 7, 9, 11
    , 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
55 -----
56 Iteration: 18
57 The best route so far is: [1, 12, 2, 13, 5, 7, 9, 11
    , 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
58 -----
59 Iteration: 19
60 The best route so far is: [1, 12, 2, 13, 5, 7, 9, 11
    , 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
61 -----
62 Iteration: 20
63 The best route so far is: [1, 12, 2, 13, 5, 7, 9, 11
    , 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
64 -----
65 Iteration: 21

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66 The best route so far is: [1, 12, 2, 13, 5, 7, 9,
    11, 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
67 -----
68 Iteration: 22
69 The best route so far is: [1, 12, 2, 13, 5, 7, 9,
    11, 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
70 -----
71 Iteration: 23
72 The best route so far is: [1, 12, 2, 13, 5, 7, 9,
    11, 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
73 -----
74 Iteration: 24
75 The best route so far is: [1, 12, 2, 13, 5, 7, 9,
    11, 3, 10, 4, 14, 6, 8, 0] with distance: 478.0
76 -----
77 Iteration: 25
78 The best route so far is: [1, 5, 13, 2, 12, 8, 6,
    14, 4, 10, 3, 11, 9, 7, 0] with distance: 507.0
79 -----
80 Iteration: 26
81 The best route so far is: [1, 5, 13, 2, 12, 8, 6,
    14, 4, 10, 3, 11, 9, 7, 0] with distance: 507.0
82 -----
83 Iteration: 27
84 The best route so far is: [1, 5, 13, 2, 12, 8, 6,
    14, 4, 10, 3, 11, 9, 7, 0] with distance: 507.0
85 -----
86 Iteration: 28
87 The best route so far is: [1, 5, 13, 2, 12, 8, 6,
    14, 4, 10, 3, 11, 9, 7, 0] with distance: 507.0
88 -----
89 Iteration: 29
90 The best route so far is: [1, 5, 13, 2, 12, 8, 6,
    14, 4, 10, 3, 11, 9, 7, 0] with distance: 507.0
91 -----
92 Iteration: 30
93 The best route so far is: [12, 6, 2, 13, 5, 7, 9,
    11, 3, 10, 4, 14, 8, 0, 1] with distance: 490.0
94 -----
95 Iteration: 31
96 The best route so far is: [2, 1, 10, 11, 13, 3, 5,
    7, 9, 6, 4, 14, 8, 0, 12] with distance: 453.0
97 -----
98 Iteration: 32
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99 The best route so far is: [2, 1, 10, 11, 13, 3, 5,
    7, 9, 6, 4, 14, 8, 0, 12] with distance: 453.0
100 -----
101 Iteration: 33
102 The best route so far is: [12, 6, 2, 13, 5, 7, 9,
    11, 3, 10, 4, 14, 8, 0, 1] with distance: 490.0
103 -----
104 Iteration: 34
105 The best route so far is: [12, 6, 2, 13, 5, 7, 9,
    11, 3, 10, 4, 14, 8, 0, 1] with distance: 490.0
106 -----
107 Iteration: 35
108 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
109 -----
110 Iteration: 36
111 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
112 -----
113 Iteration: 37
114 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
115 -----
116 Iteration: 38
117 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
118 -----
119 Iteration: 39
120 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
121 -----
122 Iteration: 40
123 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
124 -----
125 Iteration: 41
126 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
127 -----
128 Iteration: 42
129 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
130 -----
131 Iteration: 43
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132 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
133 -----
134 Iteration: 44
135 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
136 -----
137 Iteration: 45
138 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
139 -----
140 Iteration: 46
141 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
142 -----
143 Iteration: 47
144 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
145 -----
146 Iteration: 48
147 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
148 -----
149 Iteration: 49
150 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
151 -----
152 Iteration: 50
153 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
154 -----
155 Iteration: 51
156 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
157 -----
158 Iteration: 52
159 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
160 -----
161 Iteration: 53
162 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
163 -----
164 Iteration: 54

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165 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
166 -----
167 Iteration: 55
168 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
169 -----
170 Iteration: 56
171 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
172 -----
173 Iteration: 57
174 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
175 -----
176 Iteration: 58
177 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
178 -----
179 Iteration: 59
180 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
181 -----
182 Iteration: 60
183 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
184 -----
185 Iteration: 61
186 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
187 -----
188 Iteration: 62
189 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
190 -----
191 Iteration: 63
192 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
193 -----
194 Iteration: 64
195 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
196 -----
197 Iteration: 65

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198 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
199 -----
200 Iteration: 66
201 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
202 -----
203 Iteration: 67
204 The best route so far is: [6, 2, 13, 5, 7, 9, 11, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 444.0
205 -----
206 Iteration: 68
207 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
208 -----
209 Iteration: 69
210 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
211 -----
212 Iteration: 70
213 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
214 -----
215 Iteration: 71
216 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
217 -----
218 Iteration: 72
219 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
220 -----
221 Iteration: 73
222 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
223 -----
224 Iteration: 74
225 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
226 -----
227 Iteration: 75
228 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
229 -----
230 Iteration: 76

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231 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
232 -----
233 Iteration: 77
234 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
235 -----
236 Iteration: 78
237 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
238 -----
239 Iteration: 79
240 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
241 -----
242 Iteration: 80
243 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
244 -----
245 Iteration: 81
246 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
247 -----
248 Iteration: 82
249 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
250 -----
251 Iteration: 83
252 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 12, 0, 1] with distance: 436.0
253 -----
254 Iteration: 84
255 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 12, 0, 1] with distance: 436.0
256 -----
257 Iteration: 85
258 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
259 -----
260 Iteration: 86
261 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
262 -----
263 Iteration: 87

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264 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
265 -----
266 Iteration: 88
267 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
268 -----
269 Iteration: 89
270 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
271 -----
272 Iteration: 90
273 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
274 -----
275 Iteration: 91
276 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
277 -----
278 Iteration: 92
279 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
280 -----
281 Iteration: 93
282 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
283 -----
284 Iteration: 94
285 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
286 -----
287 Iteration: 95
288 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
289 -----
290 Iteration: 96
291 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
292 -----
293 Iteration: 97
294 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
295 -----
296 Iteration: 98

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297 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
298 -----
299 Iteration: 99
300 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
301 -----
302 Iteration: 100
303 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
304 -----
305 Iteration: 101
306 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
307 -----
308 Iteration: 102
309 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
310 -----
311 Iteration: 103
312 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
313 -----
314 Iteration: 104
315 The best route so far is: [13, 11, 6, 9, 2, 7, 5, 3
    , 10, 4, 14, 8, 0, 1, 12] with distance: 440.0
316 -----
317 Iteration: 105
318 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
319 -----
320 Iteration: 106
321 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
322 -----
323 Iteration: 107
324 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
325 -----
326 Iteration: 108
327 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
328 -----
329 Iteration: 109

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330 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
331 -----
332 Iteration: 110
333 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
334 -----
335 Iteration: 111
336 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
337 -----
338 Iteration: 112
339 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
340 -----
341 Iteration: 113
342 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
343 -----
344 Iteration: 114
345 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
346 -----
347 Iteration: 115
348 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
349 -----
350 Iteration: 116
351 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
352 -----
353 Iteration: 117
354 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
355 -----
356 Iteration: 118
357 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
358 -----
359 Iteration: 119
360 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
361 -----
362 Iteration: 120

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363 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
364 -----
365 Iteration: 121
366 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
367 -----
368 Iteration: 122
369 The best route so far is: [13, 11, 6, 9, 10, 3, 5,
    7, 2, 4, 14, 8, 0, 1, 12] with distance: 426.0
370 -----
371 Iteration: 123
372 The best route so far is: [13, 11, 6, 9, 12, 1, 0,
    3, 5, 7, 2, 4, 14, 8, 10] with distance: 458.0
373 -----
374 Iteration: 124
375 The best route so far is: [13, 11, 6, 9, 12, 1, 0,
    3, 5, 7, 2, 4, 14, 8, 10] with distance: 458.0
376 -----
377 Iteration: 125
378 The best route so far is: [13, 11, 6, 9, 12, 1, 0,
    3, 5, 7, 2, 4, 14, 8, 10] with distance: 458.0
379 -----
380 Iteration: 126
381 The best route so far is: [13, 11, 6, 9, 12, 1, 0,
    3, 5, 7, 2, 4, 14, 8, 10] with distance: 458.0
382 -----
383 Iteration: 127
384 The best route so far is: [13, 11, 6, 9, 12, 1, 0,
    3, 5, 7, 2, 4, 14, 8, 10] with distance: 458.0
385 -----
386 Iteration: 128
387 The best route so far is: [13, 11, 6, 9, 12, 1, 0,
    3, 5, 7, 2, 4, 14, 8, 10] with distance: 458.0
388 -----
389 Iteration: 129
390 The best route so far is: [13, 11, 6, 9, 12, 1, 0,
    3, 5, 7, 2, 4, 14, 8, 10] with distance: 458.0
391 -----
392 Iteration: 130
393 The best route so far is: [13, 11, 6, 9, 12, 1, 0,
    3, 5, 7, 2, 4, 14, 8, 10] with distance: 458.0
394 -----
395 Iteration: 131

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396 The best route so far is: [13, 11, 6, 9, 12, 1, 0,
    3, 5, 7, 2, 4, 14, 8, 10] with distance: 458.0
397 -----
398 Iteration: 132
399 The best route so far is: [13, 11, 6, 9, 12, 1, 0,
    3, 5, 7, 2, 4, 14, 8, 10] with distance: 458.0
400 -----
401 Iteration: 133
402 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
403 -----
404 Iteration: 134
405 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
406 -----
407 Iteration: 135
408 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
409 -----
410 Iteration: 136
411 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
412 -----
413 Iteration: 137
414 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
415 -----
416 Iteration: 138
417 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
418 -----
419 Iteration: 139
420 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
421 -----
422 Iteration: 140
423 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
424 -----
425 Iteration: 141
426 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
427 -----
428 Iteration: 142
```

```
429 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
430 -----
431 Iteration: 143
432 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
433 -----
434 Iteration: 144
435 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
436 -----
437 Iteration: 145
438 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
439 -----
440 Iteration: 146
441 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
442 -----
443 Iteration: 147
444 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
445 -----
446 Iteration: 148
447 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
448 -----
449 Iteration: 149
450 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
451 -----
452 Iteration: 150
453 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
454 -----
455 Iteration: 151
456 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
457 -----
458 Iteration: 152
459 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
460 -----
461 Iteration: 153
```

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462 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
463 -----
464 Iteration: 154
465 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
466 -----
467 Iteration: 155
468 The best route so far is: [13, 11, 6, 7, 12, 1, 0,
    3, 5, 9, 2, 4, 14, 8, 10] with distance: 455.0
469 -----
470 Iteration: 156
471 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
472 -----
473 Iteration: 157
474 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
475 -----
476 Iteration: 158
477 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
478 -----
479 Iteration: 159
480 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
481 -----
482 Iteration: 160
483 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
484 -----
485 Iteration: 161
486 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
487 -----
488 Iteration: 162
489 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
490 -----
491 Iteration: 163
492 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
493 -----
494 Iteration: 164

```

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495 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
496 -----
497 Iteration: 165
498 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
499 -----
500 Iteration: 166
501 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
502 -----
503 Iteration: 167
504 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
505 -----
506 Iteration: 168
507 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
508 -----
509 Iteration: 169
510 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
511 -----
512 Iteration: 170
513 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
514 -----
515 Iteration: 171
516 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
517 -----
518 Iteration: 172
519 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
520 -----
521 Iteration: 173
522 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
523 -----
524 Iteration: 174
525 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
526 -----
527 Iteration: 175

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528 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
529 -----
530 Iteration: 176
531 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
532 -----
533 Iteration: 177
534 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
535 -----
536 Iteration: 178
537 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
538 -----
539 Iteration: 179
540 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
541 -----
542 Iteration: 180
543 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
544 -----
545 Iteration: 181
546 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
547 -----
548 Iteration: 182
549 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
550 -----
551 Iteration: 183
552 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
553 -----
554 Iteration: 184
555 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
556 -----
557 Iteration: 185
558 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
559 -----
560 Iteration: 186

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561 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
562 -----
563 Iteration: 187
564 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
565 -----
566 Iteration: 188
567 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
568 -----
569 Iteration: 189
570 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
571 -----
572 Iteration: 190
573 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
574 -----
575 Iteration: 191
576 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
577 -----
578 Iteration: 192
579 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
580 -----
581 Iteration: 193
582 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
583 -----
584 Iteration: 194
585 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
586 -----
587 Iteration: 195
588 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
589 -----
590 Iteration: 196
591 The best route so far is: [13, 11, 6, 7, 12, 1, 8,
    14, 4, 2, 9, 5, 3, 0, 10] with distance: 416.0
592 -----
593 Iteration: 197

```

```
594 The best route so far is: [13, 11, 6, 7, 12, 1, 8,  
    14, 4, 2, 9, 5, 3, 0, 10]    with distance:  416.0  
595 -----  
596 Iteration:  198  
597 The best route so far is: [13, 11, 6, 7, 12, 1, 8,  
    14, 4, 2, 9, 5, 3, 0, 10]    with distance:  416.0  
598 -----  
599 Iteration:  199  
600 The best route so far is: [13, 11, 6, 7, 12, 1, 8,  
    14, 4, 2, 9, 5, 3, 10, 0]    with distance:  362.0  
601 -----  
602 End  
603  
604 Process finished with exit code 0  
605
```